Data Analyst Assessment

# Mobile Internet Adoption

## Data Preparation, Validation and Constrained Forecasting to 2030

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Implementation: R

This report documents an R-based workflow combining World Bank Internet-use data with ITU mobile Internet data. It covers extraction, harmonisation, validation, bounded forecasting, and visual communication.
Particular attention is given to indicator definitions, incomplete historical coverage, and the requirement that mobile Internet users must not exceed total Internet users.

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# 1 Summary

This assessment develops a reproducible workflow for forecasting mobile Internet users through 2030 while maintaining the logical relationship between mobile internet users and total Internet users.

The workflow uses two sources:

- World Bank: individuals using the Internet, expressed as a percentage of the population.
- International Telecommunication Union (ITU): the supplied mobile Internet-use dataset.

The World Bank series is retrieved automatically through its API. The supplied ITU file is downloaded manually and later processed in R.

The merged validation dataset contains 5,661 country–year rows. Total Internet use is missing in 769 rows, while mobile Internet use is missing in 5,430 rows. Only 209 rows contain both indicators. Consequently, the main modelling challenge is the limited and uneven coverage of the mobile series.

Validation also identifies numerous cases where the supplied mobile value exceeds the total Internet-use percentage. These discrepancies raise an important comparability issue: the two indicators may not use the same denominator.

The implemented model treats the supplied mobile values as the percentage of Internet users accessing the Internet by mobile phone. Under this interpretation, the population-based measure is constructed as:

$$\text{Mobile users (\%)} = \text{Total users (\%)} \times \frac{\text{Mobile share}}{100}$$

Forecasting combines a country-specific damped logit trend for total Internet use with a pooled logit trend for the mobile share. Their product guarantees that mobile Internet use remains below total Internet use and within the population-percentage bounds.

# 2 Objective and Scope

The objective is to forecast mobile Internet use through 2030 using a transparent and reproducible R workflow.

The assessment consists of five tasks:

1. Extract and preserve the source data.
2. Clean and harmonise country, year, and indicator values.
3. Identify missing values, unrealistic observations, and logical inconsistencies.
4. Select and justify a forecasting method that satisfies the required constraints.
5. Visualise historical estimates and forecasts clearly.

Let $T_{c,t}$ denote total Internet users as a percentage of the population in country $c$ and year $t$. Let $M_{c,t}$ denote mobile Internet users on the same population basis.

The required relationship is:

$$0 \leq M_{c,t} \leq T_{c,t} \leq 100. \tag{1}$$

The World Bank extraction requests years 2000–2025. Availability within that window varies across countries and indicators. Forecasts extend from each country's last usable mobile observation through 2030.

# 3 Task 1: Data Extraction

## 3.1 World Bank data

The World Bank indicator `IT.NET.USER.ZS` measures individuals using the Internet as a percentage of the population.

The R script retrieves the data using the World Bank API. The request specifies:

- all available economies;
- indicator `IT.NET.USER.ZS`;
- years 2000–2025;
- JSON output;
- a page size of 20,000 records.

The extraction is implemented as follows:

```
wb_url <- paste0(
  "https://api.worldbank.org/v2/country/all/",
  "indicator/IT.NET.USER.ZS",
  "?format=json&per_page=20000&date=2000:2025"
)

wb_json <- jsonlite::fromJSON(
  wb_url,
  flatten = TRUE
)

wb_raw <- wb_json[[2]]

readr::write_csv(
  wb_raw,
  "input/total_internet_raw.csv"
)
```

The returned data table is saved before cleaning. This preserves the downloaded values in a reproducible tabular form.

### 3.2 ITU data

The supplied mobile Internet dataset is imported from `mobile_internet_users.csv` and copied to:

`input/mobile_internet_raw.csv`.

This component uses a manually obtained source file rather than an automated ITU download. The World Bank API provides the automated extraction component of the workflow.

The ITU raw file is not manually edited. All transformations are performed in R after import.

### 3.3 Raw-data preservation

The workflow separates raw inputs from processed outputs:

- `input/`: source tables before cleaning;
- `output/`: cleaned data, validation reports, forecast results, and visualisations.

This separation supports traceability and allows the processing steps to be rerun without editing source observations.

## 4 Task 2: Data Cleaning and Harmonisation

### 4.1 Country identifiers

Country matching uses ISO3 codes rather than country-name strings.

For the World Bank data, the script retains ISO3 codes recognised by the `countrycode` reference list. This removes regional and income-group aggregates that do not correspond to retained country or territory codes.

For the ITU data, country names are converted to ISO3 codes. Custom mappings address naming differences such as:

- `Türkiye` to `TUR`;
- `Dominican Rep.` to `DOM`;
- `Czech Republic` to `CZE`;
- `Hong Kong, China` to `HKG`;
- `Macao, China` to `MAC`;
- `Cabo Verde` to `CPV`.

Unmatched names are printed for inspection before records without a usable ISO3 code are excluded.

### 4.2 Time format and numeric conversion

Years are converted to integers in both datasets.

The ITU table is reshaped from wide format, with a separate column for each year, into long format with one country–year observation per row.

Mobile values are processed by:

1. trimming surrounding whitespace;
2. interpreting `-` as missing;
3. removing percentage symbols;
4. converting the remaining values to numeric.

World Bank indicator values are also converted to numeric.

### 4.3 Missing and inconsistent values

Missing observations are retained during the initial cleaning stage. They are not replaced with zero or filled using cross-country averages.

The current script applies two additional rules:

- non-positive World Bank values are converted to missing;
- values above 100 are flagged before being capped at 100.

The first rule requires caution. A value of zero can be a valid percentage, so non-positive observations should not automatically be interpreted as undocumented missing-data placeholders. A stronger production implementation would preserve valid zeros, reject negative percentages, and apply source-specific missing-value rules only when supported by metadata.

The upper-bound rule also depends on the indicator definition. A cap of 100 is appropriate for a percentage of individuals but would not be appropriate for some subscription-based indicators.

### 4.4 Merging and duplicate handling

A full join on ISO3 code and year retains observations available from either source.

The main cleaned table contains:

- `country`;
- `iso3`;
- `year`;
- `total_internet_users`;
- `mobile_internet_users`.

An additional flag records whether either value exceeded 100 before capping.

The script retains one row per ISO3–year combination using `distinct()`. This prevents duplicate keys in subsequent steps, but it does not resolve conflicting duplicate values. If conflicting duplicates arise, they should be reported and reconciled explicitly rather than resolved by row order.

The cleaned merged dataset is saved as:

`output/clean_merged.csv`.

# 5 Task 3: Data Validation

## 5.1 Missing-value assessment

Table 1 summarises the exported missing-value report.

Table 1: Missing-value summary for the cleaned merged dataset

| Validation measure | Number of rows |
|---|---|
| Total country–year rows | 5,661 |
| Total Internet-use value missing | 769 |
| Mobile Internet-use value missing | 5,430 |
| Both indicators missing | 747 |
| Total available, mobile missing | 4,683 |
| Both indicators available | 209 |

The counts describe the full merged country–year panel, not only the countries or years with mobile observations. Therefore, the large number of missing mobile values reflects differences in coverage as well as gaps within individual series.

The limited overlap motivates a parsimonious pooled model rather than a separate complex time-series model for every country.

Country-level mobile coverage is saved separately in:

`output/validation_mobile_coverage.csv`.

## 5.2 Range checks

The workflow checks for:

- negative percentages;
- values exceeding 100 before capping;
- missing observations.

The available file `validation_out_of_range.csv` contains column headers but no flagged rows.

This result refers to the exported validation run and the implemented checks. It does not verify that the indicators have equivalent definitions or denominators.

### 5.3 Temporal consistency and outlier checks

The script implements three diagnostic rules:

1. Total Internet-use changes exceeding 15 percentage points per year, adjusting for the gap between available observations.
2. Mobile observations more than 15 percentage points from the median of the same country's other mobile observations, where at least three observations are available.
3. Mobile declines exceeding 10 percentage points between successive available observations.

The third check records the gap between observations but does not annualise the decline. It should therefore be interpreted as a between-observation decline diagnostic, not always a year-on-year change.

The exported mobile-outlier file contains ten flagged observations across Czechia, Egypt, Israel, Japan, Latvia, Poland, and Russia.

These flags identify observations for review; they do not establish that the source values are incorrect.

### 5.4 Constraint violations

For rows containing both indicators, the script checks:

$$M_{c,t} > T_{c,t}.$$

Several discrepancies are too large to be explained by minor rounding differences. Selected examples are presented in Table 2.

Table 2: Selected discrepancies before denominator adjustment

| Country | Year | Total | Supplied mobile | Gap |
|---|---|---|---|---|
| Malawi | 2023 | 17.99 | 98.10 | 80.11 |
| Zimbabwe | 2020 | 29.30 | 97.42 | 68.12 |
| Indonesia | 2019 | 47.69 | 96.95 | 49.26 |
| Lithuania | 2023 | 88.50 | 93.75 | 5.24 |

Notes: Values are rounded to two decimal places. Gap is the supplied mobile value minus the total Internet-use percentage, expressed in percentage points. The mobile column is not assumed to have a verified population denominator at this stage.

The full list is retained in:

`output/validation_constraint_violations.csv`.

## 6 Task 4: Modelling and Forecasting

### 6.1 Modelling strategy

The forecast decomposes mobile Internet use into:

1. the total population share using the Internet;
2. the share of Internet users accessing it by mobile phone.

The final forecast is:

$$\widehat{M}_{c,t} = \widehat{T}_{c,t}\widehat{S}_{c,t}. \tag{2}$$

This approach is chosen for three reasons:

- it represents the required relationship directly;
- logit transformations keep the components bounded;
- pooling the mobile trend reduces the need to estimate separate trends from very short country series.

The method is deliberately parsimonious and consistent with the assessment's limited time budget.

## 6.2 Total Internet-use path

### 6.2.1 Historical interpolation

For each country, observed total Internet-use values are retained. Missing years between the first and last observations are filled using linear interpolation.

Interpolation is used to construct an aligned historical path. It is not interpreted as a newly observed survey value.

The output distinguishes total values as:

- `observed`;
- `interpolated`;
- `forecast`.

### 6.2.2 Logit transformation

For modelling, the total percentage is converted to a proportion and clipped away from zero and one:

$$p_{c,t} = \min\left\{0.999,\, \max\left(0.001,\, \frac{T_{c,t}}{100}\right)\right\}.$$

The transformed series is:

$$z_{c,t} = \operatorname{logit}(p_{c,t}) = \log\left(\frac{p_{c,t}}{1 - p_{c,t}}\right). \tag{3}$$

### 6.2.3 Recent country-specific trend

The script estimates a linear trend using observations from the last ten calendar years ending at the country's final observed year:

$$z_{c,t} = a_c + b_c t + \varepsilon_{c,t}.$$

At least four observations are required to estimate the trend. Otherwise, the slope is set to zero.

The fitted total-use slope is restricted to be non-negative:

$$b_c^+ = \max(\widehat{b}_c, 0).$$

This is a modelling assumption that rules out a declining future total-use path. It is not a property guaranteed by the data.

#### 6.2.4 Damped forecast

Let $t_c^*$ denote the final observed total-use year and let $h$ denote the forecast horizon.

The damped accumulation factor is:

$$D(h) = \sum_{j=1}^{h} \phi^j = \frac{\phi(1-\phi^h)}{1-\phi}, \qquad \phi = 0.8. \tag{4}$$

The total-use forecast is:

$$\widehat{T}_{c,t_c^*+h} = 100 \text{ logit}^{-1}\left(z_{c,t_c^*} + b_c^+ D(h)\right). \tag{5}$$

Damping makes future increments progressively smaller rather than extending the recent trend indefinitely.

### 6.3 Mobile-share preparation

Under the conditional denominator interpretation in Section **??**, the supplied ITU value is divided by 100 to obtain $S_{c,t}$.

The modelling step applies a separate isolated-outlier rule. An interior observation is excluded when it differs by more than 0.15 from the linearly interpolated value implied by the remaining observations for that country.

This rule differs from the median-based validation diagnostic: the validation step flags observations for review, whereas the modelling step removes observations meeting its isolated-spike rule.

### 6.4 Pooled model for the mobile share

Before applying the logit transformation, mobile shares are clipped to the interval $[0.01, 0.995]$:

$$\widetilde{S}_{c,t} = \min\{0.995, \max(0.01, S_{c,t})\}.$$

The pooled model is:

$$y_{c,t} = \alpha_c + \beta(t - 2018) + u_{c,t}, \tag{6}$$

where:

$$y_{c,t} = \text{logit}(\widetilde{S}_{c,t}).$$

Country fixed effects $\alpha_c$ allow different average levels, while the common slope $\beta$ summarises the time trend across the available country series.

The model is estimated in R using:

```
fit <- lm(
  y ~ factor(iso3) + t,
  data = fit_df
)
```

The common mobile-share slope is not constrained to be positive.

### 6.5 Forecast output

The combined historical-estimate and forecast table is saved as:

`output/mobile_internet_forecasts.csv`.

Its principal fields are shown in Table 3.

Table 3: Forecast output structure

| Field | Description |
|---|---|
| `country` | Standardised country name |
| `iso3` | ISO3 country code |
| `year` | Calendar year |
| `type` | Historical estimate or forecast |
| `total_internet_users` | Total-use path, % population |
| `mobile_share_of_users` | Mobile share, expressed as a percentage |
| `mobile_internet_users` | Reconstructed or forecast mobile use, % population |
| `mobile_lower80` | Approximate lower mobile-use band |
| `mobile_upper80` | Approximate upper mobile-use band |
| `total_source` | Observed, interpolated, or forecast total-use value |

Forecast coverage is limited to countries with retained mobile observations and a usable aligned total-use path. The workflow does not generate unsupported mobile forecasts for every country in the World Bank dataset.

### 6.6 Statistical model summary:

Table 4: Interpretation of the pooled logit-share model summary

| Statistic | Reported value | Interpretation |
|---|---|---|
| Residual standard error | 0.6359 | Estimated residual standard deviation on the logit scale. This is not an error measured in population percentage points. |
| Residual degrees of freedom | 158 | Number of observations minus the rank of the fitted model, representing the degrees of freedom available for estimating residual variance. |
| Multiple $R^2$ | 0.8304 | Country fixed effects and the common time trend jointly explain 83.04% of the in-sample variation in the logit-transformed mobile-use share. |
| Adjusted $R^2$ | 0.7541 | Reported goodness-of-fit measure after accounting for the number of estimated predictors relative to the sample size. |
| Overall F-statistic | $F(71, 158) = 10.89$ | Tests whether all non-intercept coefficients, including country effects and the common time trend, are jointly zero. |
| Overall p-value | $< 2.2 \times 10^{-16}$ | Strong evidence against the joint null hypothesis under the usual regression assumptions. This does not establish that the time trend is individually significant. |

## 7 Task 5: Visualisation

### 7.1 Overview

The graph presents historical estimates and forecasts of mobile Internet use as a percentage of the population. Black solid lines represent the historical estimates, constructed by combining the ITU mobile-use share with the World Bank total Internet-use percentage. Red dashed lines show forecasts from each country's final retained mobile observation through 2030.

The historical trajectories generally indicates the increasing of mobile Internet use, although both adoption levels and the pace of change differ considerably across countries, because of expensive bandwith price and geographical difficulities. Although Several trajectories approach the upper end of the percentage scale, while others remain substantially lower.

The forecasts suggest that mobile internet user increases for most countries, with many trajectories becoming progressively flatter. This pattern is consistent with the model specification: the logit transformation bounds the forecasts, while trend damping reduces the size of future increments. Consequently, the apparent movement towards saturation should be interpreted partly as a modeling outcome rather than as a directly observed future development.

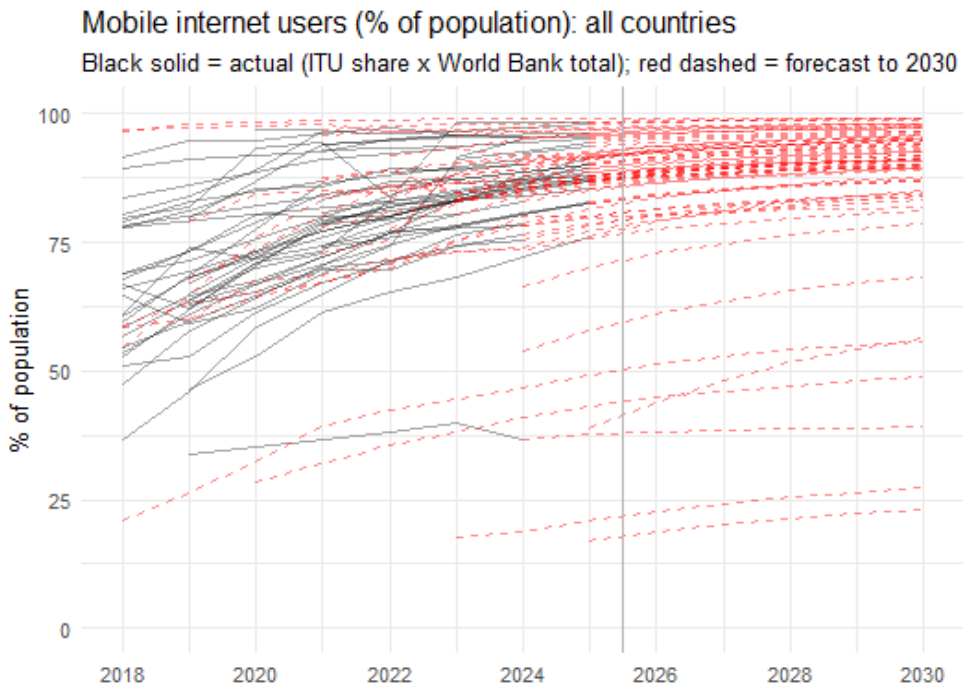



Figure 1: Forecasted Internet Users Over Time (2000–2030)

## 7.2 Individual Country Forecasts

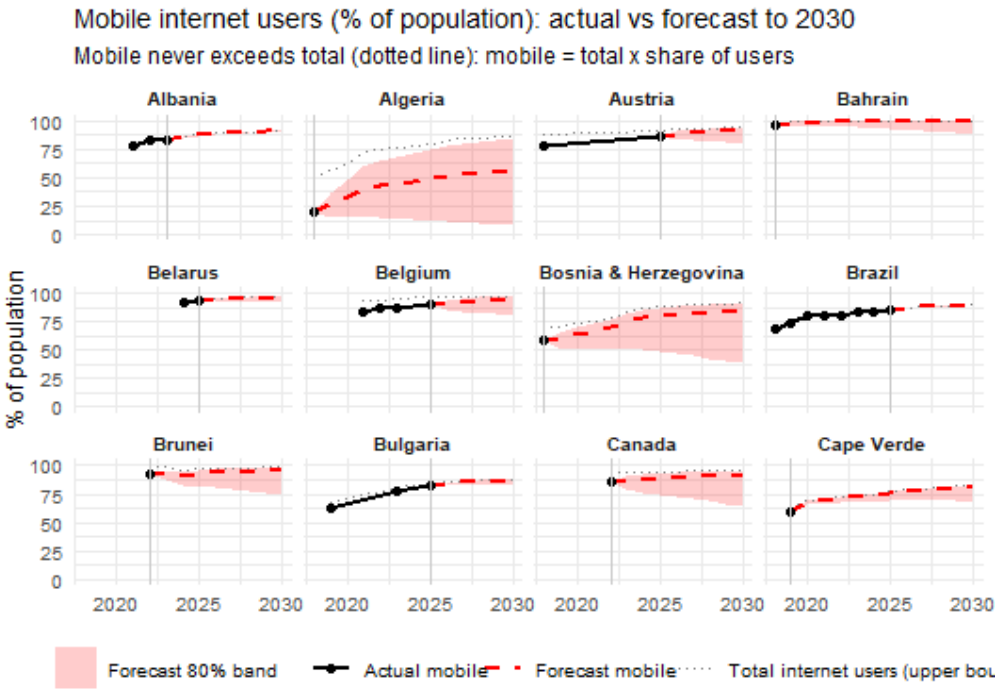



Figure 2: Historical estimates and forecasts of mobile Internet use through 2030: country panels, group 1.

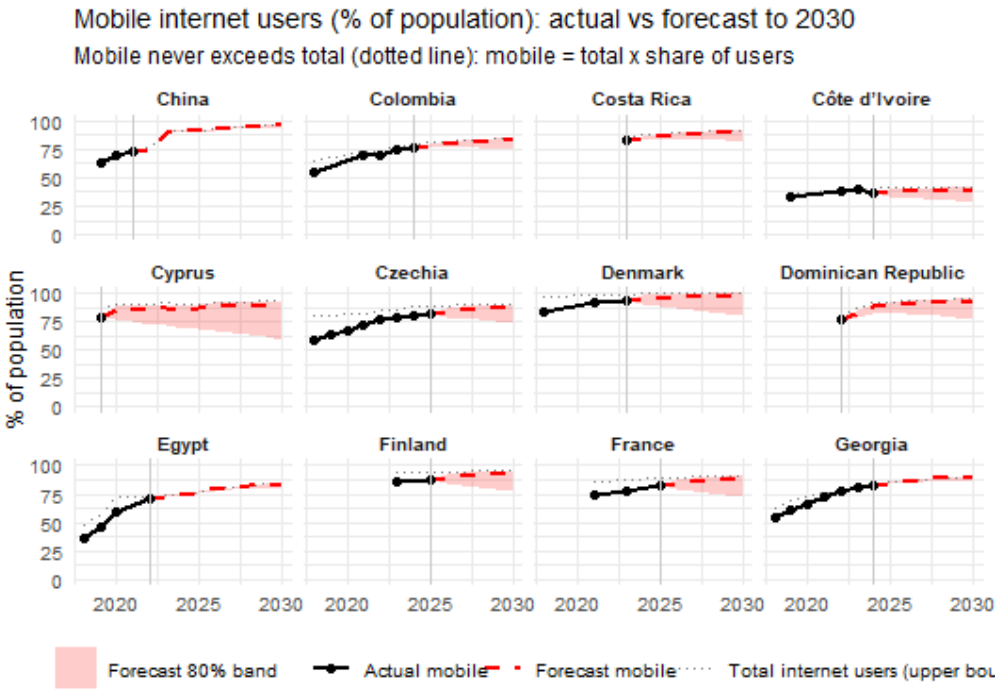



Figure 3: Historical estimates and forecasts of mobile Internet use through 2030: country panels, group 2.

### 7.3 Discussion of Country-Level Forecasts

### 7.4 Export

The charts are exported as a multipage PDF:

`output/plots.pdf`.

The export uses landscape A4-like dimensions of 11.7 by 8.3 inches.

The plotting stage should read the newly exported forecast table, not an older file left over from an earlier run.

## 8 Findings and Interpretation

### 8.1 Data comparability is the central issue

The most important validation finding is the mismatch between the supplied mobile values and total Internet-use percentages.

Large discrepancies show that mechanically treating both columns as equivalent population percentages is not defensible without checking their definitions.

The product-based reconstruction is coherent only if the mobile series is confirmed to represent a share of Internet users.

### 8.2 Mobile coverage limits model complexity

Only 209 rows contain both indicators in the merged validation table.

This limited overlap supports a simple model that pools trend information rather than estimating a highly parameterised process for each country.

However, pooling introduces its own restriction: countries with different adoption dynamics share the same mobile-share trend.

### 8.3 Forecasts reflect two adoption processes

The product formulation separates:

- growth in overall Internet access;
- changes in the use of mobile phones among Internet users.

When the mobile share is already close to one, future population-based mobile growth is driven mainly by the total Internet-use path.

When the mobile share is lower, both components can materially affect the result.

## 9 Limitations and Potential Improvements

- Big countries like USA, India and Australia has no data of mobile internet using in the data set.
- Countries contribute different numbers of observations and have different final observation years. As a result, forecast horizons and the strength of country-specific evidence differ.
- The pooled model imposes a common logit-scale trend. It may not capture country-specific policy changes, infrastructure constraints, or saturation patterns.
- A possible extension is a hierarchical model with partially pooled country slopes, subject to sufficient data.
- The damping parameter $\phi = 0.8$ is selected rather than estimated.

  A sensitivity analysis could compare alternative damping values and report how much the 2030 forecasts change.
- Negative fitted trends in total Internet use are set to zero. This simplifies the long-run adoption story but prevents the model from overfitting or underfitting.
- Simulation or bootstrap methods could be used to show more efficient forecast result.

## 10 Reproducibility and Deliverables

### 10.1 Software

The workflow is implemented in R using:

- `jsonlite` for API JSON processing;
- `readr` for CSV input and output;
- `dplyr` and `tidyr` for data manipulation;
- `stringr` for text cleaning;
- `countrycode` for country harmonisation;
- `purrr` for functional programming utilities;
- `ggplot2` for visualisation.

The analytical report has been written by latex code.

### 10.2 Project structure

```
project/
|-- main.R
|-- input/
|   |-- total_internet_raw.csv
|   `-- mobile_internet_raw.csv
`-- output/
    |-- clean_merged.csv
    |-- clean_validated.csv
    |-- validation_missing_summary.csv
    |-- validation_mobile_coverage.csv
    |-- validation_out_of_range.csv
    |-- validation_mobile_outliers.csv
    |-- validation_constraint_violations.csv
    |-- mobile_internet_forecasts.csv
    `-- plots.pdf
```

### 10.3 Submission checklist

Before submission:

1. Confirm the ITU indicator definition and denominator.
2. Review the treatment of zero-valued World Bank observations.
3. Add the forecast CSV export before the plotting stage.
4. Rerun `main.R` from a clean R session.
5. Check that expected countries have forecasts through 2030.
6. Confirm that all required forecast values are non-missing.
7. Verify the population bounds and mobile–total constraint.
8. Regenerate the charts from the newly exported CSV.
9. Use accurate labels for reconstructed historical estimates and approximate uncertainty bands.

10. Include the required raw inputs, R script, forecast CSV, and chart PDF.

## 11 Conclusion

The workflow demonstrates how extraction, cleaning , validation and forecasting can be combined in a transparent R pipeline. Its principal methodological feature is the decomposition of population-based mobile Internet use into total Internet use and a bounded mobile share. The workflow provides a reproducible basis for producing mobile Internet-use forecasts through 2030. The lack of infrastructure in some countries is slowing them down in the race for internet availability.

## 12 Github

https://github.com/Jackie556/Euromonitor.git

## Data Sources

1. World Bank Open Data. Individuals using the Internet (% of population), indicator `IT.NET.USER.ZS`. https://data.worldbank.org/indicator/IT.NET.USER.ZS
2. International Telecommunication Union. ICT Statistics and the supplied mobile Internet-use export. https://www.itu.int/en/ITU-D/Statistics/Pages/stat/default.aspx
3. Internal assessment materials: assignment instructions, `main.R`, cleaned datasets, and exported validation reports.
4. Perplexity AI has been used to make this analytical report.